

Svrops.com - Windows XP Unattended Install

43–54 minutes

Introduction

The following are the procedures to follow for an **unattended install of Windows XP Pro with Office XP Professional** from a Windows 2000 Server. There are various ways to script an unattended install of a Microsoft operating system, and this is just one possible installation scenario.

If you are looking to implement an unattended install of Windows XP, this document will get you started. The various steps can be modified to suit your environment.

Note: Although this document discusses Windows XP and Office XP, the same procedures can be followed for Windows 2000 and Office 2000.

Terms

Installation Point(s)	The folder(s) on a Windows 2000 server that house the Windows XP operating system files and/or the Office XP files. Also called Distribution Points.
Source Server	The server that houses the files necessary for the installation. The installation point(s) will be on the source server.
Target PC	The PC that you will be installing Windows XP and Office XP on.
Unattended Install	The automated installation and configuration of an operating system and applications using a script to answer installation questions. Ideally, no user intervention or input should be needed once the installation starts.

Assumptions

It is assumed you have some experience with unattended installs, DOS batch files, Office XP installations, Windows XP and a general understanding of networking and PC installs. This document does not attempt to go into every detail about Windows and Office installations.

Description of the Target PC

The target PC in this installation is fully Windows XP compliant. Ideally, Windows XP will have all the drivers necessary for the hardware on your PC(s). Since XP is a relatively new operating system, this should not be a problem, unless you are using bleeding edge hardware. If this is the case, and you do have to add drivers, this is a fairly simple process. And though I will not be covering that topic in this document, you can find the information on how to do this in the Windows 2000 [unattend.doc](#) file (it is the same for Windows XP) or in the Windows XP [deployment help files](#). Additionally, the following MS Knowledgebase article describes the process:

<http://support.microsoft.com/default.aspx?scid=kb;en-us;Q254078>

Make/Model Dell Dimension 4100
Processor Pentium III - 1.0GHz

Memory	256MB of PC133 SDRAM
Hard Drive	20GB ATA 100 hard drive connected to the onboard hard drive Controller.
Network Card	3COM 3c905TX running at 100 Mbps/Full duplex, connected to a 3COM 10/100 autosensing switch over CAT5 UTP cable.
Sound Card	Sound Blaster Live Value
Video Card	ATI Rage 128 Pro AGP 4x
Monitor	17" Plug and Play CRT - Dell branded monitor
Peripherals	1.44MB Floppy drive, Samsung 24x CD ROM Drive
<u>Basic Procedures</u> 	

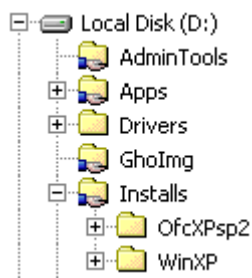
The basic procedures for this installation are as follows. All steps will be automated (except for creating the Installation Points). In this installation, the installer will only have 2 things to do: Put the boot disk in the target PC, reboot the PC.
Enter the name of the computer during the XP Pro install.

1. [Create Installation Points](#) for Windows XP Pro and Office XP on a Windows 2000 server.
2. [Create an unattend.txt file](#) that will be the script file for the unattended install.
3. [Create a Network boot disk](#) that will boot up the target PC, connect to and login to a Windows 2000 Domain, map a drive to a share on the source server and start the unattended installation.
4. [Prepare the target PC](#) by deleting the partitions on the target PC's hard drive and create a FAT16, 2GB partition. This will be the C: drive.
5. Once the hard drive has been partitioned and formatted as FAT16, reboot the PC, reconnect to the source server and start installing Windows XP Pro.
6. [The XP installation will start](#), files will be copied, the hard drive will be formatted as NTFS, and the boot/system partition (C: drive) will be increased from 2GB to 6GB.
During the XP Pro install, you will be prompted for a computer name. Enter a unique computer name. When the install finishes, the PC will reboot and the local Administrator will automatically log in.
7. [Once the Administrator logs in](#), a batch file will kick off that will do the following:
 - Modify some registry settings on the target PC.
 - Copy the Windows XP I386 directory from the source server to the C:\I386 directory on the target PC.
8.
 - Create and format as NTFS, a second partition which will be the E: drive. The C: drive will be the boot/system partition, and D: will be the CD ROM drive.
 - Install Office XP on the E: drive under E:\Program Files.
 - Delete the Default User profile and copy a preconfigured Default User profile to the target PC.
 - Do a silent install of Recovery Console.
 - Reboot the PC for a final time. User logs in.

Creating the Installation Points

On your Windows 2000 server create a folder called **Installs**. This should be at the root of a drive. In this example it will be the D: drive. Under Installs create the following folders:

OfcXPsp2
WinXP



Next, on the Windows 2000 server/domain, create a user named **Installman**. This will be the username that is used to login to the domain and do the unattended install. Share the Installs folder. Give Authenticated Users Full Control Share Permissions. For the NTFS Permissions, give Administrators Full Control and for Installman give Read&Execute, List Folder Contents and Read. No other users need to be added to the permissions list.

Note: How you set up user and group access will depend on your Domain standards. The important thing is to give only the minimum permissions necessary.

Windows XP Pro Installation Point

To create the Windows XP Installation Point, simply copy the contents of your XP Pro CD to the D:\Installs\Winxp folder. This will create the installation point. Next, you should slipstream the latest **service pack** into the Winxp folder so you have an integrated install of Windows XP SPx. For instructions on slipstreaming see [this page](#) →.

Note: You could just copy the I386 folder from the CD to \Winxp, but then you wouldn't be able to slipstream service packs into your XP installation point.

Office XP Installation Point

To create the Office XP Installation Point, put the Office CD in the server CD-ROM drive or a workstation CD-ROM drive and run **setup.exe /a**. (You will need an Enterprise version of Office XP to create an Administrative Installation Point) Your destination folder will be \servername\installs\ofcxsp2. With Office, you cannot just copy the contents of the CD to the server.

For more detailed information about creating an Office Administrative Installation Point see this MS web page:

<http://www.microsoft.com/office/ork/xp/one/depb01.htm>

And, like the Windows XP installation point, you should slipstream the latest Office XP **service pack** into your installation point. This process is different than slipstreaming for Windows XP. You can find information about applying Office XP SP2 to an Office installation point at this MS web site:

<http://www.microsoft.com/office/ork/xp/journ/Oxpsp2a.htm>

Or at this web site [here](#) →

Additional Windows XP Folders

Once you have your base Installation Points created, you will add subfolders to the D:\installs\winxp folder. Create these folders:

d:\installs\winxp\boot

d:\installs\winxp\boot\flag

Note:

As part of the target PC's hard disk preparation, a file called flag.txt will be added to and later deleted from the d:\installs\winxp\boot\flag folder. Because of this, you will need to add the **modify permission** for the user Installman, for the d:\installs\winxp\boot\flag folder on the source server.

This will allow Installman to create and delete the flag.txt file. The \boot folder is the folder that will be accessed when the target PC logs in to the Domain to start the unattended install. Inside this folder is the unattend.bat file, the unattend.txt file and files used to partition and format the target PC hard drive. This will be discussed later.

In addition to the boot folder, create the following folders under d:\installs\winxp\I386:

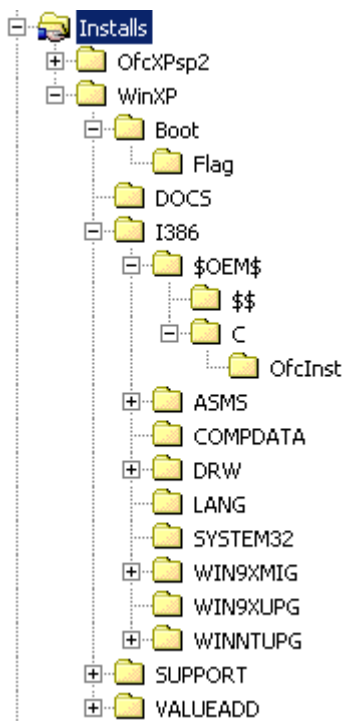
d:\installs\winxp\I386\SOEM\$

d:\installs\winxp\I386\SOEM\$\\$

d:\installs\winxp\I386\SOEM\$\C

d:\installs\winxp\I386\SOEM\C\OfcInst

Your directory structure should look like this. **Note:** If you would like to download a .zip file that has the \SOEM\$ folder and all it's files, as well as the \boot folder and all it's files, you can do so [here](#).



The following table lists the \I386 subfolders that can be used by an unattended installation and what each folder is used for. Basically, if you need to copy any files to the target PC during an unattended install or will be using a cmdlines.txt file, you create an \$OEM\$ folder and subfolders under the \I386 folder on your Windows XP installation point. The \$OEM\$ folder and it's subfolders have specific purposes which are described in the table.

For our installation scenario we will only need some of the possible folders. Those folders and their contents will be explained later in this document.

Note: When using the \$OEM\$ folder and it's subfolders to copy files to the target PC, you have to include the **OemPreinstall=Yes** line in the unattend.txt file under the [Unattended] section.

\\$OEM\$	Holds all folders and files for an unattended installation. Also, Cmdlines.txt, and files it references, are put in this folder.
\\$OEM\$\Textmode	Contains updated mass storage drivers and HAL files required during the text-mode portion of Setup.
\\$OEM\$__\$	Contains files that Setup copies to the %Windir% (e.g., C:\winnt or C:\Windows) folder during installation.
\\$OEM\$__\$\Help	Contains custom Help files that Setup copies to the %Windir%\Help folder during installation.
\\$OEM\$__\$\System32	Contains custom Help files that Setup copies to the %Windir%\System32 folder during installation.
\\$OEM\$__\$1	Represents the root of the drive on which you install WinXP (usually C:\) and contains any files that Setup should copy to the boot partition during installation.
\\$OEM\$\\$1\Pnpdrvs	Contains new or updated PnP drivers. The user specifies the folder name in the unattend.txt file used during unattended installations. For example, this folder might be named \\$OEM\$\\$1\Pnpdrvs.
\\$OEM\$\\$1\SysPrep	Contains files that Setup uses for the Sysprep-based method of installation.
\\$OEM\$\\$drive_letter	Represents the root of a particular volume on the system (e.g., \\$OEM\$\C represents the C: drive) and contains any files that WinXP Setup should copy to this partition during installation.
\\$OEM\$\\$drive_letter\subfolder	Represents a particular subfolder of the drive (e.g., \\$OEM\$\C\MyFolder) and contains any files that Setup should copy to the subfolder during installation. Multiple instances of this type of folder can exist under the \\$OEM\$\\$drive_letter folder.

The Unattend.txt file

Below is the unattend.txt file used for this installation. I will not go into a detailed explanation of what each heading and value is for. Most of it is self explanatory. For a complete description of the unattend.txt file see the Windows 2000 [unattend.doc](#) file or the Windows XP [deployment help files](#).

What I have done, is added in

red text, an explanation of some of the more important values and what they are for.

Note: This installation is using a volume license version of Windows XP Pro which does not require activation. If you are using a version of Windows XP that does require **activation**, add the following to the unattend.txt file:

Under the [Unattended] section

AutoActivate=Yes

Under the [UserData] section, make sure you put your Product ID (serial number)

ProductID=yourProductID

see this MS KB article for more info:

[http://support.microsoft.com/default.aspx?scid=kb;\[LN\];Q291997](http://support.microsoft.com/default.aspx?scid=kb;[LN];Q291997)

Creating an unattend.txt file

The unattend.txt file is the script file used to answer all of the questions that are asked during a normal Windows XP installation. It is nothing more than a simple text file that can be created in Notepad. It doesn't have to be named unattend.txt. You could call it winxp.txt or whatever you want. It does, however, require a .txt extension.

Microsoft offers you some help in creating the unattend.txt file with a utility called **Setupmgr.exe**. This program can be found on the XP Pro CD under Support\Tools in the Deploy.cab file. Setup Manager will walk you through creating an unattend.txt file. However, it doesn't support every aspect of unattend.txt so you might have to edit the file in Notepad after using Setup Manager. If necessary, you can download the XP Setup Manager [here](#).

Note: [TechProGuid](#) has a nice step by step how to article for the WinXP Setup Manager.

```
;SetupMgrTag
```

```
[Data]
```

```
AutoPartition=1
```

```
MsDosInitiated="0"
```

```
UnattendedInstall="Yes"
```

```
[Unattended]
```

```
UnattendMode=FullUnattended
```

```
OemSkipEula=Yes
```

Setting OemPreInstall=Yes is necessary if you create an \$OEM\$ directory under \i386 to copy files to the target PC and/or are using a cmdlines.txt file. If you aren't copying files to the target PC or using a cmdlines.txt file, then set it to No

```
OemPreinstall=Yes
```

```
TargetPath=\WINDOWS
```

The 2 lines below are used to convert the file system to NTFS and then extend the boot/system partition an extra 4GB.

```
FileSystem=ConvertNTFS
```

```
ExtendOEMPartition=4000
```

```
[GuiUnattended]
```

```
AdminPassword="happyday"
```

```
EncryptedAdminPassword=NO
```

```
OEMSkipRegional=1
```

```
TimeZone=35
```

```
OemSkipWelcome=1
```

The following 2 lines tell setup to login 1 time automatically after the installation has finished. Setup will only use the local Administrator account to login

AutoLogon=Yes
AutoLogonCount=1

[UserData]

You could add a ComputerName= line in this section. However, if you are installing multiple PCs, and since each PC needs a unique computer name, leaving out this line causes Setup to prompt you for a computer name. Alternatively, you could use a UDF file to provide unique user information.

ProductID=xxxxx-xxxxx-xxxxx-xxxxx-xxxxx
FullName="Jane Doe"
OrgName="Widgets"

[Display]

BitsPerPel=32
Xresolution=1024
YResolution=768
Vrefresh=75

[Components]

This section is where you tell XP to not install certain components. The line below tells XP not to install MSN Explorer.
msnexplr=off

[TapiLocation]

CountryCode=1
Dialing=Tone
AreaCode=305

[RegionalSettings]

LanguageGroup=1
Language=00000409

[Branding]

BrandIEUsingUnattended=Yes

[URL]

Home_Page=www.svrops.com

[Proxy]

Proxy_Enable=0
Use_Same_Proxy=1

[Identification]

This section joins your PC to the domain indicated, creates a computer account and uses the username and password indicated to create the account.
JoinDomain=MYDOMAIN.com

CreateComputerAccountInDomain=Yes
DomainAdmin=msmith
DomainAdminPassword=password

[Networking]

This section and the ones to follow were generated by Setup Manager.
They describe what networking protocols and components to install.

InstallDefaultComponents=No

[NetAdapters]

Adapter1=params.Adapter1

[params.Adapter1]

INFID=*

[NetClients]

MS_MSClient=params.MS_MSClient

[NetServices]

MS_SERVER=params.MS_SERVER

[NetProtocols]

MS_TCPIP=params.MS_TCPIP

[params.MS_TCPIP]

DNS=Yes

UseDomainNameDevolution=No

EnableLMHosts=Yes

AdapterSections=params.MS_TCPIP.Adapter1

[params.MS_TCPIP.Adapter1]

SpecificTo=Adapter1

DHCP=Yes

WINS=No

NetBIOSOptions=0

[GuiRunOnce]

This section tells setup to run the indicated commands at the first logon after setup finishes. You can use Autologon in [GuiUnattended] above to login and automatically run these commands. For multiple commands just add more lines - Command1=, Command2=, etc.

Command0=c:\ofcinst\ofcxp.bat

Creating the Network boot disk

Creating the **Network boot disk** that will boot your target PC and connect it to the Windows 2000 domain could be the hardest part of the unattended install. If you tried to do it from scratch, you might never get it done. Microsoft, in Windows NT Server, had a utility that would help you create a DOS bootable disk that would connect a computer to a NT Domain, but this program was left out of

Windows 2000 Server. Not that the program was very good to begin with. It was somewhat crude and Microsoft never developed it after it's initial release.

Luckily, I can provide you with a [floppy disk image](#) from which you can create the network boot disk. The image file is self extracting and will create a disk that will boot the PC using DOS 6.22, connect to a Windows NT or 2000 domain, login, and map a drive to a share on the source server. The disk has support for 3COM's 3c90x line of network adapters, and changing or adding network card support is an easy process using a batch file. Read the Readme.txt on the floppy for instructions.

Another possibility, if you have access to **Ghost** Enterprise or Ghost 2003, is to use Ghost to create a network boot disk. Ghost makes the process very easy and supports many network cards.

The network boot disk, for this installation scenario, does **3 things**. It connects the PC to a domain, maps a drive to a share on the source server and starts the unattended install. Nothing else is done on the floppy.

Note: Once XP setup starts copying files to your target PC's hard drive, **remove** the network boot disk from the PC. It will no longer be needed.

The rest of this unattended installation is done from the Win2000 server. That is what we will be talking about next.

The **autoexec.bat** file on the Network Boot Disk:

```
@echo off
:: This file first checks to see if you want to change your Protocol.ini and
:: System.ini files before you connect to the domain. If you do, then it calls
:: netcard.bat to change these files then returns to this file to connect
:: to the network. Make sure you copy your DOS network card driver to a:\net
:: first, if you are going to change the .ini files.

cls
path=a:\;a:\net

echo.
echo.
echo          ** Change Ini Files? **
echo.
echo  Do you need to modify the Protocol.ini and System.ini files? [Y/N]
echo.

a:\yn.com
if not errorlevel 1 goto START
if errorlevel 1 if not errorlevel 0 goto INIFILES

:INIFILES
@call a:\netcard.bat
goto START
```

:START

cls

a:\net\smartdrv.exe /q

a:\net\net initialize

a:\net\netbind.com

a:\net\umb.com

a:\net\tcptsr.exe

a:\net\tinyrfc.exe

a:\net\nmtsr.exe

a:\net\emsbfr.exe

a:\net\net start

:: The following lines can be deleted or modified to suit your needs.

:: They can be used to map a drive to a server and start an unattended install.

:: Note: You can only map a drive to the root of a share when using DOS. DOS

:: doesn't support deep redirection so you can't map to a folder under

:: a share, e.g. \\srv2k01\installs is the share. You can't map to \\srv2k01\installs\winxp.

net use I: \\SRV2K01\Installs

@echo off

::Check to see if the drive exists

if not exist I:\winxp*.* goto NODRIVE

:: Start the unattended install with unattend.bat

I:

cd\

cd winxp\boot

unattend.bat

:NODRIVE

cls

echo.

echo The mapped drive - I: was not found. Please try

echo remapping the drive or reboot the PC and try again.

goto end

:END

a:

cd\

Preparing the Target PC 

Earlier, when you were creating your Windows XP Pro [installation point](#), you created a winxp\boot folder. This folder is what the network boot disk will connect the target PC to. In this folder are the following files and folder:



Note: If you would like to download a .zip file that has the **\\$OEM\$** folder and all it's files, as well as the **\boot** folder and all it's files, you can do so [here](#).

Once you are connected to the d:\installs\winxp\boot directory, the unattend.bat file (below) is run. Unattend.bat calls **newpart.bat** which uses **gdisk.exe** to partition and format the hard drive. Gdisk.exe is a souped up version of the DOS Fdisk utility. It is made by Symantec and comes with Norton Ghost. After the hard drive is partitioned and formatted (FAT16), the PC must be rebooted. This is done by calling **reboot.com**. The target PC reboots, and again connects to the source server and looks for the flag.txt file in the \boot\flag directory. If this file is found, then the PC's hard drive has been formatted and the batch program jumps down to start the Windows XP Pro install using **unattend.txt**.

The **yn.com** file is a simple file that waits for you to press Y or N. It uses an errorlevel to check which was pressed. See the YN.txt file for info.

The **unattend.bat** file:

```
@echo off
::Batch file to kick off XP Pro Unattended Install
::The I: Drive, which is used below, is created on the
::floppy disk that connects the PC to the server
::with the net use command, eg. net use i: \\server01\installs.
::Once the WinXP Pro install starts copying files, you can
::remove the floppy disk from the PC.

cls
:: Set environmental variables
set AnswerFile=i:\winxp\boot\unattend.txt
set SetupFiles=i:\winxp\i386

:: After Gdisk partitions and formats the hard drive the PC must be
:: rebooted. The following line checks to see if the PC was rebooted
:: by checking for the flag.txt file.
:: If the file exists, the PC was formatted and partitioned, and the
:: batch files jumps down to install WinXP.

if exist flag\flag.txt goto INSTALL

echo.
echo Your PC is connected to the Network, and
echo Windows XP Pro Setup will begin.
```

```

echo.
echo Your Hard Drive will be partitioned and
echo formatted.
echo.
echo Do you want to continue (Y/N)?
echo.

yn.com
if not errorlevel 1 goto STOP
goto PART

:PART
:: Call Norton Gdisk.exe to partition and format the hard drive, then reboot the PC.
@call newpart.bat

:: This line creates the flag.txt file. If the flag.txt file
:: exists in the \Flag directory, then the PC was partitioned and formatted.
echo Gdisk was Run > flag\flag.txt

:: Reboot.com is a small program that reboots a PC
reboot.com
goto END

:INSTALL
If exist flag\flag.txt del flag\flag.txt
::Start the WinXP Pro Install
I:
cd\
winxp\i386\winnt.exe /u:%AnswerFile% /s:%SetupFiles%
goto END

:STOP
cls
echo.
echo User Terminated Install
echo.
goto END

:END

```

The XP Pro Installation Starts

The target PC's hard drive has been partitioned, formatted and XP Pro is being installed. XP setup will format the hard drive as **NTFS** and extend the partition to **6GB**. You will be asked to enter a unique computer name at the beginning of the XP install, and the computer will be added to the network. As part of the XP Pro unattended install, the files and/or directories under the **d:\installs\winxp\I386\SOEM\$** folder on the source server will be copied to the target PC hard drive. A file, shutdown.exe,

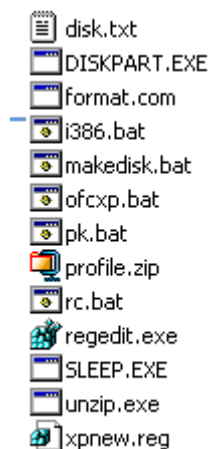
which is in the d:\installs\winxp\I386\\${OEM}\$ folder on the source server, will be copied to C:\windows. The directory, Ofcinst and it's files, which is in the d:\installs\winxp\I386\\${OEM}\$C folder on the source server, will be copied to the root of C:\.

About 20 mins. later, the install will finish on it's own and the target PC will be rebooted. At this point, the XP Pro installation is completed on the target PC. The rest of the unattended install is for customizing the target PC after XP Pro has been installed.

After the PC reboots, the local Administrator user will automatically login, and the final phase of the unattended install will start.

Run the ofcxp.bat file in GuiRunOnce 

Once the Administrator logs in, a batch file, **ofcxp.bat**, which is in the c:\ofcinst folder, will kick off and finish the install. The c:\ofcinst folder on the target PC contains the following files:



It is worth mentioning here that instead of calling a batch file from **GuiRunOnce**, you could use the **cmdlines.txt** file to do many of the same things. Cmdlines.txt is just a text file you can make in Notepad with a line for each action you want run (I included a sample cmdlines.txt in the \${OEM}\$ directory download). Cmdlines.txt is called at the end of setup and before the PC reboots for the last time. You put cmdlines.txt in the I386\\${OEM}\$ directory of your installation point along with any files that cmdlines will be calling or referencing.

The only drawback to using cmdlines is you **can't** see what is happening as the commands are run. If there is a problem, it is more difficult to ascertain the source of the problem. Also, I had problems modifying the registry and installing Office using cmdlines.txt (they just didn't work right), and that is one reason I chose not to use it. It might be different for you, so it is worth testing if you would rather use it.

The **ofcxp.bat** file does the following:

1. Modify some registry settings on the target PC.

This is done by the **xpnew.reg** file

2. Copy the Windows XP I386 directory from the source server to the C:\I386 directory on the target PC.

The d:\installs\winxp\I386\ files and folders on the source server are copied to the c:\I386 folder on the target PC by the **i386.bat** file. This file copies the files then checks for success.

Note: The I386 folder on the source server also has the \$OEM\$ folder in it which isn't needed on the target PC. The \$OEM\$ folder, in c:\I386, is deleted after all the files and folders are copied.

3. Create and format as NTFS, a second partition which will be the E: drive. The C: drive will be the boot/system partition, and D: will be the CD ROM drive.

This is done by calling **makedisk.bat**. The makedisk.bat file uses the Microsoft **Diskpart** utility.

Diskpart can create partitions from within Windows. It is essentially an automated Disk Manager. The file **disk.txt** has the commands that Diskpart.exe will use to create the new partition. A new partition - E: will be created and the remaining drive space will be used for the E: drive.

Once Diskpart creates the partition, the WinXP **Format.com** program formats the partition as NTFS.

4. Install Office XP on the E: drive under E:\Program Files.

Now that the E: drive has been created, an E:\Program Files directory is created and the automated install for Office XP is started. This is done with a standard Office install command using a MST file to configure the Office installation. You can get more information on Office XP deployment at this [MS web site](#).

5. Delete the Default User profile and copy a preconfigured Default User profile to the target PC.

The reason for this step is to create a **default user environment** for each user that logs on to the PC. If you set up a reference PC to test your unattended install on, you can create a local account called Testuser, login as Testuser, configure the Desktop, Internet Explorer settings, Windows Explorer settings, Display settings and so on. These settings, which are saved in the Testuser's profile, can be copied to other users by copying the Testuser profile to the Default User profile. Then, each time a new user logs in for the first time, they will get those settings from the Default User profile.

For this install, once I configured my Testuser's settings and copied the Testuser profile to the Default User profile, I used a zip utility, Winzip, and zipped up the Default User profile directory. I then copied that .zip file to the source server. During the unattended install of XP, that .zip file, **profile.zip**, gets copied to the target PC's hard drive under C:\ofcinst. In the ofcxp.bat file, **PK.bat** gets called and deletes the Default User directory and subdirectories on the target PC. Then, using an unzip program, **unzip.exe**, that will run in an XP command prompt window, it unzips the profile.zip file and the Default User directory and subdirectories get created.

6. Do a silent install of Recovery Console.

This is fairly straight forward. Since the XP I386 directory has been copied to the target PC (see step 2 above), you can install **Recovery Console** by issuing the command:

c:\i386\winnt32.exe /cmdcons /unattend. The /unattend switch allows Recovery Console to install without prompting you with the usual questions.

The **RC.bat** file, in C:\ofcinst, installs Recovery Console.

7. Reboot the PC for a final time. User logs in.

Finally, we are through with the Windows XP install. It is time to reboot the PC. You can do this with the MS Shutdown.exe utility. During the unattended install, shutdown.exe was copied to C:\windows. It is called from the ofcxp.bat file.

Note: Although Shutdown.exe was copied to the target PC from the source server, this was done to show how you would copy a file to the system root folder. However, it really isn't necessary to copy this file since Windows XP Pro comes with Shutdown.exe in the Windows\system32 folder.

8. Oh wait...There is one **last thing to do**. Now that the install is finished, we no longer need the C:

\ofcinst folder and it's files. The very last thing to do is delete this folder. This is done with the commands:

```
cd\
```

```
Start /b %ComSpec% /cRD /s /q "%~dp0"
```

The **ofcxp.bat** file:

```
@echo off
```

```
c:
```

```
cd\
```

```
:: Create folders on target PC
```

```
if not exist c:\logfiles\CON md Logfiles
```

```
if not exist c:\I386\CON md I386
```

```
if not exist c:\Logfiles\CON goto END
```

```
if not exist c:\I386\CON goto END
```

```
cd\
```

```
:: Create an authenticated connection to the server by mapping a drive.
```

```
net use q: \\srv2k01\installs /user:arbor01\installman /persistent:no catfish
```

```
:: Changes some registry settings
```

```
cd ofcinst
```

```
start /wait regedit /s xpnew.reg > c:\logfiles\xpnew.txt
```

```
:: Copy the i386 files to c:\I386 on the target PC
```

```
start /wait i386.bat
```

```
:: Create E: Drive on the target PC
```

```
c:
```

```
cd\
```

```
cd ofcinst
```

```
start /wait makedisk.bat
```

```
if not exist e:\CON goto BADRIVE
```

```
if exist e:\CON md "e:\Program Files"
```

```
if not exist "e:\Program Files" goto END
```

```
:: Start the automated install of Office XP.
```

```
start /wait \\srv2k01\installs\OfcXPsp2\setup.exe TRANSFORMS="\"
```

```
\srv2k01\Installs\OfcXPsp2\OffXP2e.MST" /qb /lew c:\logfiles\officexp.txt
```

```
:: Recreate the default user profile on the target PC
```

```
start /wait pk.bat
```

```
:: Install Recovery Console on the target PC
```

```
echo.
```

```
echo Installing Recovery Console...
```

```
start /wait rc.bat
```

```

echo.
if not exist c:\cmdcons\CON echo Recovery Console Not Installed.
if exist c:\cmdcons\CON echo Recovery Console Installed.

:: Reboot the PC using the Shutdown program located in the c:\windows folder.
start /wait %windir%\shutdown.exe /L /T:4 "Computer will now Reboot" /C /R /Y
goto EXIT

:END
echo Folder(s) not created.
echo Office XP was not installed.
goto STOP

:BADRIVE
echo E: drive does not exist
goto STOP

:EXIT
:: The following lines delete this file and the directory it's in.
cd\
Start /b %ComSpec% /cRD /s /q "%~dp0"
exit

:STOP
pause

```

Final Thoughts

Creating and testing a Windows unattended install can be **time consuming** and frustrating, depending on how complex you try to make it. There is no definitive document/book/web site that outlines how to create a Windows unattended install. Information and documentation for unattended installs is **piecemeal** and scattered all over the place. Even Microsoft's own documentation on unattended installs is like this. You will have to read books, search web sites, experiment, and ask questions. Also, there is no definitive way of implementing an unattended install because every environment is different. If you plan on adding multiple applications to your automated installation, an unattended install might not be the best idea. Disk imaging might be a better choice.

Disk imaging software has matured in the last couple of years, and creating an image of a reference PC and deploying that image is probably the fastest way to set up PCs. Using Sysprep with the Disk Image allows for some variety in your PC hardware, however; the more homogenous your hardware environment, the better off you will be (standardization is key).

There are also specialized 3rd party tools that can be used for operating systems deployment, but these tools tend to be rather expensive.

Whatever your choice for operating systems deployment, unattended installs are still, and will be for the near future, a viable way of deploying Microsoft operating systems.

Hopefully this document gave you some ideas and direction in creating an unattended install of Windows XP with Office XP.